

Transaction User Insights Report

Data period: 2024-03-01 to 2024-05-31

Generated on: 2026-10-02

1. Executive Summary

- 278,932 transactions from 24,321 users across 6,489 merchants were processed, covering 2024-03-01 to 2024-05-31.
- Users split into 5 behavioral segments; the largest is 'Bargain Hunters' at 45.8% of users. Chapter 4 profiles every segment and Chapter 6.3 breaks down each supplementary business question by segment.
- 'Food & Dining' is the best-selling category by transaction count; 'Entertainment' is the least ordered.
- The recommended minimum Android version to cover 90% of active Android users is 1.0.
- Loyalty members transact 11.43 times on average vs. 11.48 for non-members (not statistically significant ($p = 0.4617 \geq \alpha = 0.05$)).
- Transaction volume rises sharply during the Eid al-Fitr window (2024-04-07 to 2024-04-13), the strongest single pattern in the data.
- This dataset is synthetic: most cross-variable relationships tested are weak or not statistically significant. Chapter 5 (assumptions & limitations) covers this in detail.

2. Data Audit

The dataset contains 278,932 transactions, 24,321 users, 6,489 merchants and 21 MCC categories. This chapter is kept brief; it exists to establish how much to trust the features built in Chapter 3.

Transaction status breakdown

Status	Count
completed	175,955
pending	93,584
failed	9,393

Key limitations

- Data is synthetic: 93,584 transactions (33.6%) have a blank status (neither completed nor failed) and correspondingly blank rating/refund/discount/notes/loyalty fields; these are treated as a third 'pending' category rather than as errors.
- 100% of transactions with a rating are on failed transactions (9,393 of 9,393 failed transactions carry a rating) - the opposite of what would be expected operationally, and as if the rating and status fields were generated independently.
- The administrative boundary file (GADM, 394 regencies/cities) predates several splits; there are 514 regencies/cities today. This has no effect on the 9 cities present in this dataset.
- No promo-source column exists, so merchant- vs platform-funded discounts are inferred heuristically from each merchant's discount rate vs. the platform baseline.

3. Feature Engineering: Methodology, Assumptions & Findings

24,321 users, 97 engineered features per user, built in four stages: transaction-level derivation, per-user aggregation, Big Five proxies, and demographic proxies. Each section below states the assumption behind a feature group and a live finding computed from this run's data, not a static description.

3.1 Activity (recency, frequency, regularity)

Assumption: how often and how recently a user transacts, and how regular that rhythm is (burstiness), reflects engagement independent of how much they spend.

- Median active months: 3.0 of the 3-month window.
- Median tenure (first-to-last transaction): 77 days.
- Median burstiness: -0.07 (negative = regular spacing, positive = bursty).

3.2 Value (spend level and variability)

Assumption: average spend and its coefficient of variation (CV) capture both how much a user spends and how consistent that spend is, which average alone would hide.

- Median avg. net amount per purchase: 34,627.
- Median coefficient of variation: 0.79.

3.3 Category & merchant diversity

Assumption: Shannon entropy over a user's MCC/merchant distribution measures variety-seeking independent of how many transactions they have, unlike a raw count of distinct categories.

- Median category entropy: 1.56.
- Median merchant repeat ratio: 0.0% of purchases revisit a previously used merchant.

3.4 Geography

Assumption: a user's dominant transaction city approximates their home base. Finding: geo activity in this dataset is scattered rather than concentrated around one city for most users (see 3.9 for how this affected the work-location proxy).

- 99.1% of users transact in more than one city.
- Median share of transactions in a user's single most common city: 50.0%.

3.5 Payment & promo behavior

Assumption: payment-method mix and discount usage reflect financial behavior, not identity.

- Balance-only users: 10.1%; credit-card-only: 0.0%.
- 54.6% of users have used a promo at least once.

3.6 Quality (failure, refund)

Assumption: failure and refund rates, computed per user, surface quality issues a merchant-level view would average away.

- Median per-user failure rate: 0.0%.
- Median per-user refund rate: 0.0%.

3.7 Time patterns

Assumption: weekend share and payday-window share capture discretionary vs. routine spending timing.

- Median weekend share: 18.2%.
- Median payday-window share: 18.2%.

3.8 Device

Assumption: OS family and version are observed directly from user_agent, no inference needed.

- 66.4% of users are primarily on Android.
- 57.4% were seen at least once on a legacy OS version.

3.9 Big Five Personality Score

Methodology: mean z-score of 3-4 signed behavioral indicators per trait (table below), converted to a 0-100 percentile (src/features/personality.py).

Trait	Indicators (sign)
Openness	category_entropy (+), n_categories (+), merchant_diversity (+)
Conscientiousness	burstiness (-), failure_rate (-), essential_share (+), payday_share (-)
Extraversion	social_share (+), weekend_share (+), n_cities (+)
Agreeableness	avg_rating (+), merchant_repeat_ratio (+), notes_share (-)
Neuroticism	refund_rate (+), rating_std (+), cv_net_amount (+)

Assumption: this indicator-to-trait mapping is this project's own hypothesis (ordinary consumer-behavior reasoning, e.g. shopping across more categories suggests openness), not copied from a validated instrument. It is grounded in a real research precedent for the general idea, not a specific cited method: Gladstone, Matz & Lemaire (2019, Psychological Science) linked real UK bank transactions to a self-reported Big Five questionnaire (BFI-10) and found Extraversion most positively correlated with dining/drinking spend, which matches this project's social_share indicator for the same trait; the other traits here use categories that study's data had and this dataset's 21 MCC codes do not, so there was nothing to compare for those. Ramon, Matz, Farrokhnia & Martens (2021) found Conscientiousness and Neuroticism most predictable from spending data and Openness/Agreeableness least predictable; Neuroticism is one of two traits that pass the reliability check below, which is at least directionally consistent.

Trait	Cronbach's alpha	Mean inter-item corr.	Reliable (alpha >= 0.5)
openness	0.9896	0.9693	Yes
conscientiousness	-0.4431	-0.0831	No
extraversion	0.4131	0.19	No
agreeableness	-0.0376	-0.0122	No
neuroticism	0.7617	0.5159	Yes

Finding: Cronbach's alpha only checks whether a trait's own indicators move together in this dataset; it cannot confirm the score matches a real person's personality, since no real personality label exists here to check against. A trait with alpha below 0.5 should not be interpreted as a real personality measurement; its indicators do not move together enough to justify combining them. Both studies cited above validated against a real self-report questionnaire the account holder filled in, not a third-party rater's impression; that is the validation path this project would need (e.g. a short BFI-10 or TIPI survey at onboarding), not crowd annotation, since transaction history gives an outside observer nothing to watch the way a video would.

3.10 Demographic Proxies

Every demographic attribute below gets at least one data-driven proxy attempt, with the assumption behind it stated explicitly, except Industry of Employment, which is documented as infeasible after an explicit attempt (no field ties a user to an employer at all). Every proxy ships a confidence column so a weak signal is visible rather than hidden.

Big Five Personality Score (1st priority) - columns: openness_score, conscientiousness_score, extraversion_score, agreeableness_score, neuroticism_score

- Methodology: Mean z-score of 3-4 signed behavioral indicators per trait, converted to a 0-100 percentile.
- Assumption: Spending behavior reflects personality the way it does in published transaction-based studies (Gladstone, Matz & Lemaire, 2019, Psychological Science); indicator-to-trait direction is this project's own hypothesis, not copied from a validated instrument.
- Finding: See Cronbach's alpha table below for per-trait reliability.

Gender (1st priority) - columns: gender_lean_signal, gender_lean_confidence

- Methodology: Composite z-score of spend share in a small set of stereotype-associated retail categories (e.g. beauty/clothing vs. fuel/sporting goods/hardware).
- Assumption: Retail-category stereotypes correlate with gender. This is a weak societal stereotype, not a validated or ethical ground truth - reported as an exploratory 'lean', never an identity claim.
- Finding: 29.5% of users have a non-'none' confidence signal.

Age (1st priority) - columns: age_bracket_estimate, age_bracket_confidence

- Methodology: Reuses the student/parental MCC signals: university spend -> 18-24 bracket; school spend -> 25-45 (parent) bracket; otherwise unknown.
- Assumption: Only users who shop at a university or pay a school-related merchant are age-informative; everyone else is genuinely unknown, not guessed.
- Finding: 12.0% of users have a non-'none' confidence signal.

Income Level (1st priority) - columns: income_proxy_tier

- Methodology: Combines credit-card usage share with the user's percentile rank of average net spend.
- Assumption: Credit-card access and higher average spend both correlate loosely with income.
- Finding: medium: 96.1%, low: 3.5%, high: 0.4%

Educational Background (1st priority) - columns: education_signal, education_confidence

- Methodology: Presence of university-category (MCC 8220) spend.
- Assumption: Currently studying at a university is the only educational signal transaction data can support; attained education level (e.g. high-school vs. bachelor's) cannot be inferred.
- Finding: 5.9% of users have a non-'none' confidence signal.

Home Location (1st priority) - columns: home_city_estimate, home_city_confidence

- Methodology: The user's most frequent transaction city overall; confidence scales with how dominant that city is in the user's activity.
- Assumption: A user's most common transaction city approximates where they live.

- Finding: 100.0% of users have a non-'none' confidence signal.

Work Location (1st priority) - columns: work_location_estimate, work_location_confidence

- Methodology: The user's most frequent city on weekdays only, when it differs from the overall (home) dominant city.
- Assumption: No time-of-day field exists (only transaction date), so a weekday/weekend split is used instead of the usual daytime-hours work-cluster approach.
- Finding: 66.7% of users have a non-'none' confidence signal.

Working Status (2nd priority) - columns: working_status_estimate, working_status_confidence

- Methodology: payday_share above a threshold combined with weekend_share below a threshold.
- Assumption: Spending concentrated around common pay dates and weekday-heavy activity loosely suggests regular salaried income; confidence is capped at 'low' because neither input is a direct employment signal.
- Finding: 27.4% of users have a non-'none' confidence signal.

Industry of Employment (2nd priority) - columns: (not estimated)

- Methodology: Attempted: tested whether a user's own spending-category concentration could proxy their employer's industry; rejected as conflating consumption category with employer sector.
- Assumption: No field in this dataset ties a user to an employer or occupation at all.
- Finding: Not estimated - see methodology column.

Health Status (2nd priority) - columns: health_care_signal_count, health_care_signal_confidence

- Methodology: Count of pharmacy (MCC 5912) purchases vs. the chance baseline rate for that category.
- Assumption: Pharmacy visits loosely indicate health-related spending; says nothing about condition or severity.
- Finding: 41.9% of users have a non-'none' confidence signal.

Homeownership Status (2nd priority) - columns: homeowner_signal_count, homeowner_signal_confidence

- Methodology: Count of hardware/furniture (MCC 5251, 5712) purchases vs. chance baseline.
- Assumption: Home-improvement spend loosely suggests owning (vs. renting) a home.
- Finding: 26.5% of users have a non-'none' confidence signal.

Marital Status (2nd priority) - columns: marital_family_proxy, marital_family_confidence

- Methodology: Composite: has a parental signal AND is a loyalty member AND has a geographically settled primary city (primary_city_share >= 0.5).
- Assumption: Three independently weak signals stacked together; confidence is capped at 'low' because none of the three is a direct marital-status signal.
- Finding: 0.9% of users have a non-'none' confidence signal.

Parental Status (2nd priority) - columns: parental_signal_count, parental_signal_confidence

- Methodology: Count of school-related (MCC 8211) purchases vs. chance baseline.

- Assumption: Paying a school-related merchant loosely suggests having a school-age child.
- Finding: 6.8% of users have a non-'none' confidence signal.

Vehicle ownership (2nd priority) - columns: vehicle_signal_count, vehicle_signal_confidence

- Methodology: Count of fuel/service-station (MCC 5541) purchases vs. chance baseline.
- Assumption: Buying fuel loosely suggests owning a motor vehicle.
- Finding: 25.5% of users have a non-'none' confidence signal.

4. User Segmentation: Methodology & Segment Stories

Users were grouped into behavioral segments with K-Means (k=5, chosen by silhouette score across k=3-6). Clustering used only behavioral features (recency, activity, value, diversity, discount/loyalty usage, payment mix, quality); geography, device, payment category, personality scores, and the demographic proxies from Chapter 3.10 were deliberately excluded from the clustering inputs and used only to write the persona stories below, so a segment is not simply a relabeled existing column.

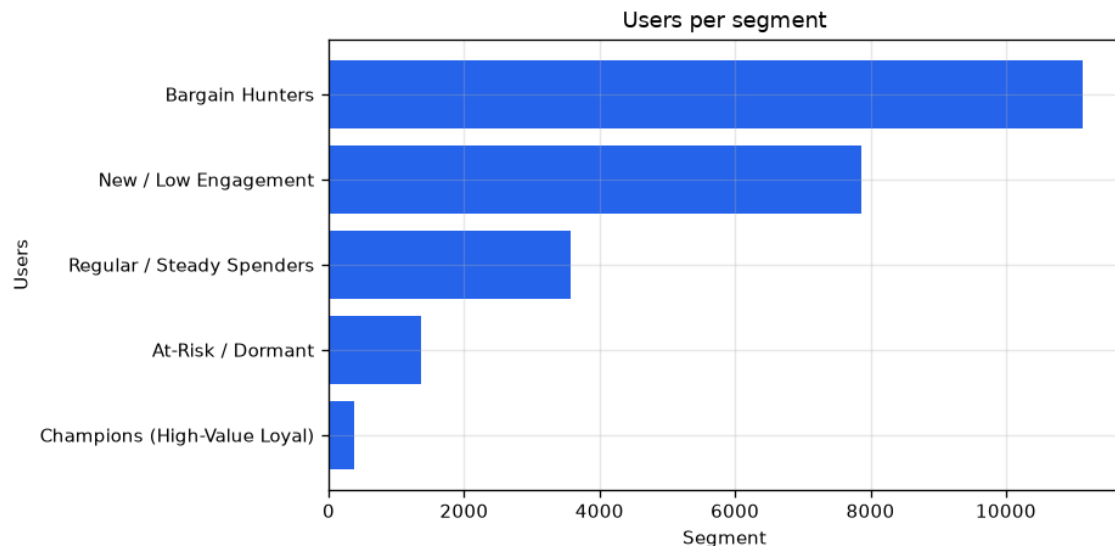
Why not plain RFM

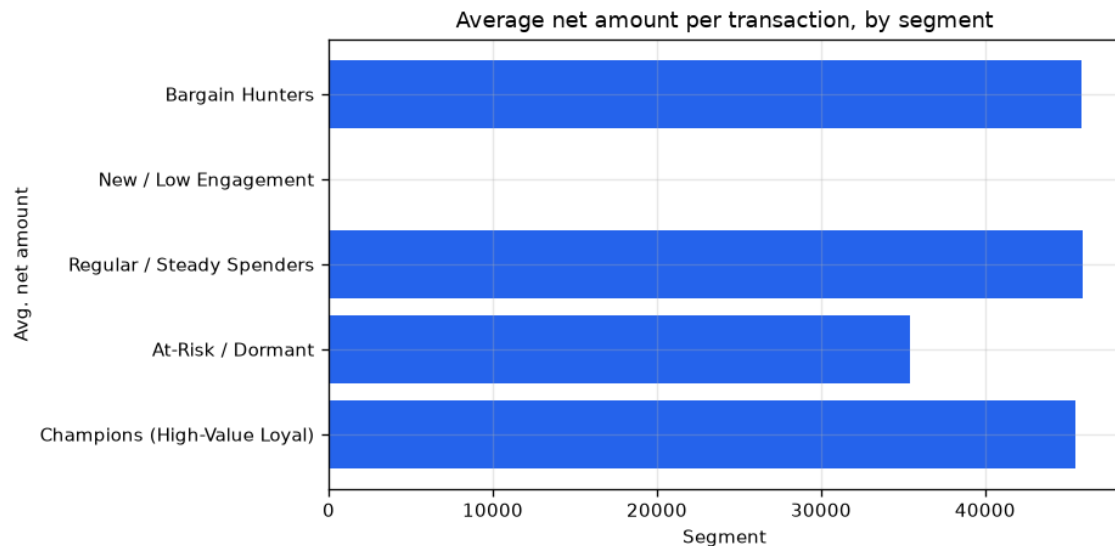
This segmentation is deliberately designed to go beyond traditional RFM. The 15 clustering inputs here include recency/frequency/monetary (recency_days, n_valid_purchases, avg_net_amount) but also diversity (category_entropy, merchant_diversity), promo/loyalty behavior (discount_tx_share, loyalty_member), payment mix (credit_card_share), quality (failure_rate, refund_rate), and time patterns (weekend_share, lebaran_lift), so segments can separate on behavior that RFM alone would miss entirely.

k selection (silhouette score)

k	Inertia	Silhouette
3	194720.3	0.3478
4	176328.6	0.3577
5	155397.7	0.3658
6	145104.4	0.3234

Segment sizes and value





Segment profiles & stories

Bargain Hunters - 11,134 users (45.8% of users).

- Recency: 7 days since last transaction; active in 3.0 months on average.
- 10.5 valid purchases, avg. net amount 45,867.
- Discount usage: 17.0% of purchases; credit card share: 20.1%; loyalty member rate: 0.0%.

Story: Typically based in Kota Jakarta Pusat; medium-income proxy; mixed payment; stands out for: currently-studying signal (1.6x the overall rate), 18-24 student-leaning age signal (1.6x the overall rate); averages 10.5 purchases at 45,867 each.

New / Low Engagement - 7,871 users (32.4% of users).

- Recency: 7 days since last transaction; active in 3.0 months on average.
- 0.0 valid purchases, avg. net amount 3.
- Discount usage: 0.0% of purchases; credit card share: 20.0%; loyalty member rate: 0.0%.

Story: Typically based in Kota Jakarta Pusat; medium-income proxy; mixed payment; stands out for: female-leaning retail signal (1.5x the overall rate); averages 0.0 purchases at 3 each.

Regular / Steady Spenders - 3,574 users (14.7% of users).

- Recency: 7 days since last transaction; active in 3.0 months on average.
- 10.5 valid purchases, avg. net amount 45,942.
- Discount usage: 16.9% of purchases; credit card share: 20.3%; loyalty member rate: 100.0%.

Story: Typically based in Kota Jakarta Pusat; medium-income proxy; mixed payment; stands out for: family/parental signal (6.3x the overall rate), regular-income (payday) signal (1.5x the overall rate); averages 10.5 purchases at 45,942 each.

At-Risk / Dormant - 1,369 users (5.6% of users).

- Recency: 25 days since last transaction; active in 2.0 months on average.
- 5.6 valid purchases, avg. net amount 35,406.

- Discount usage: 13.3% of purchases; credit card share: 20.0%; loyalty member rate: 15.3%.

Story: Typically based in Kota Jakarta Pusat; medium-income proxy; mixed payment; no demographic proxy is notably over-represented vs. the overall population; averages 5.6 purchases at 35,406 each.

Champions (High-Value Loyal) - 373 users (1.5% of users).

- Recency: 7 days since last transaction; active in 3.0 months on average.

- 12.3 valid purchases, avg. net amount 45,461.

- Discount usage: 17.4% of purchases; credit card share: 20.1%; loyalty member rate: 22.8%.

Story: Typically based in Kota Surabaya; medium-income proxy; mixed payment; stands out for: family/parental signal (1.7x the overall rate), male-leaning retail signal (1.7x the overall rate); averages 12.3 purchases at 45,461 each.

5. Assumptions & Limitations

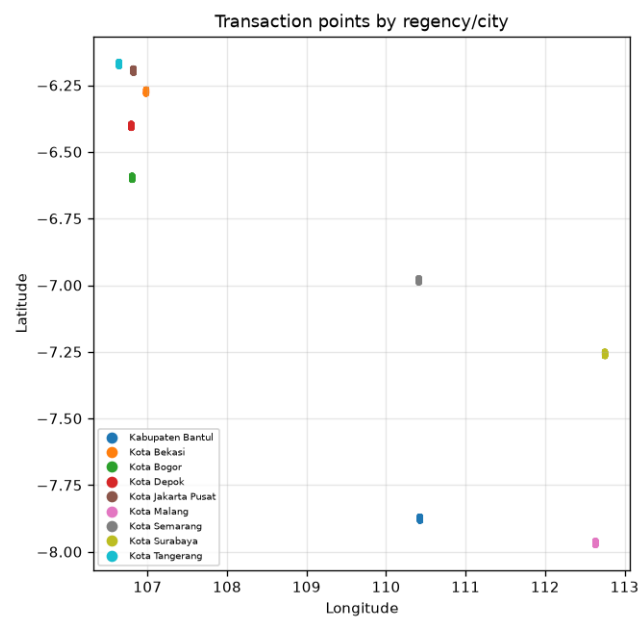
- transaction_amount is assumed to be the gross amount (before discount); net_amount = transaction_amount - promo_amount.
- MDR cost assumptions: balance = 0.70%, credit_card = 2.20% of transaction value.
- Eid al-Fitr period: 2024-04-07 to 2024-04-13.
- Every requested demographic attribute except Industry of Employment has a heuristic proxy (Chapter 3.10); confidence is mostly 'low' or 'none' by design because the underlying signals are weak. None of them should be treated as a validated classification - see demographic_feasibility.csv for the one attribute left undone and why.
- For real-world use, the biggest gaps are: a genuine promo-source field, a resolved status for the 33.6% of transactions currently blank, and self-reported ground truth (e.g. a short BFI-10 or TIPI survey) to actually validate the Big Five and demographic proxies against real labels.

6. Appendix: Supplementary Analysis

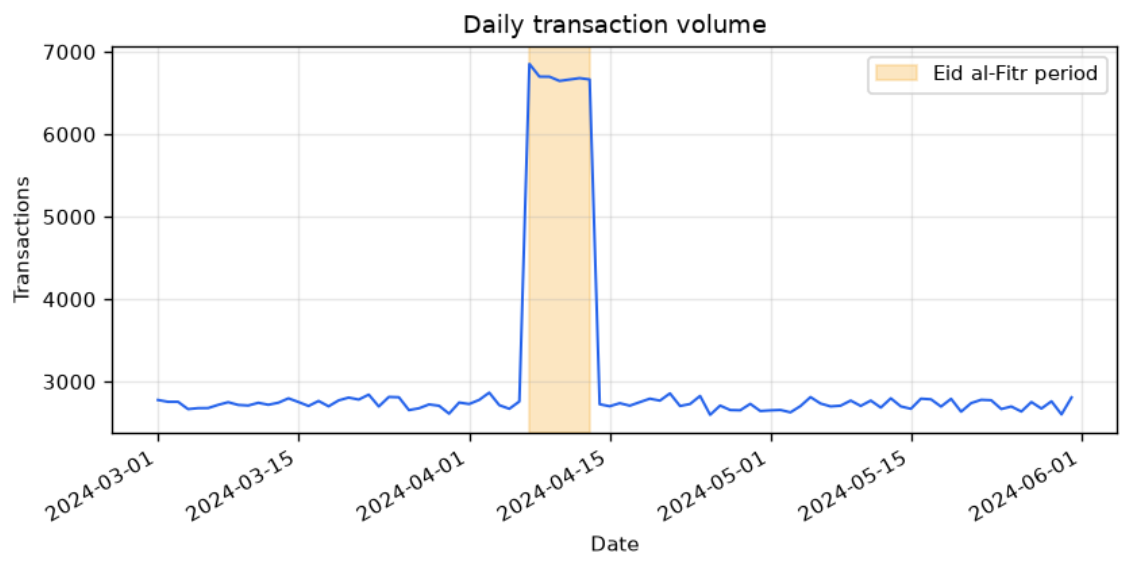
Everything in this appendix goes beyond the core feature engineering and segmentation deliverables in the chapters above. It demonstrates additional exploration of the same data: regional/temporal patterns and eight concrete business questions, each with a statistical test and a per-segment breakdown using the segmentation from Chapter 4.

6.1 Regional Profile

Geo-matching coverage: 100.0% of transactions matched to a regency/city.



6.2 Time Patterns



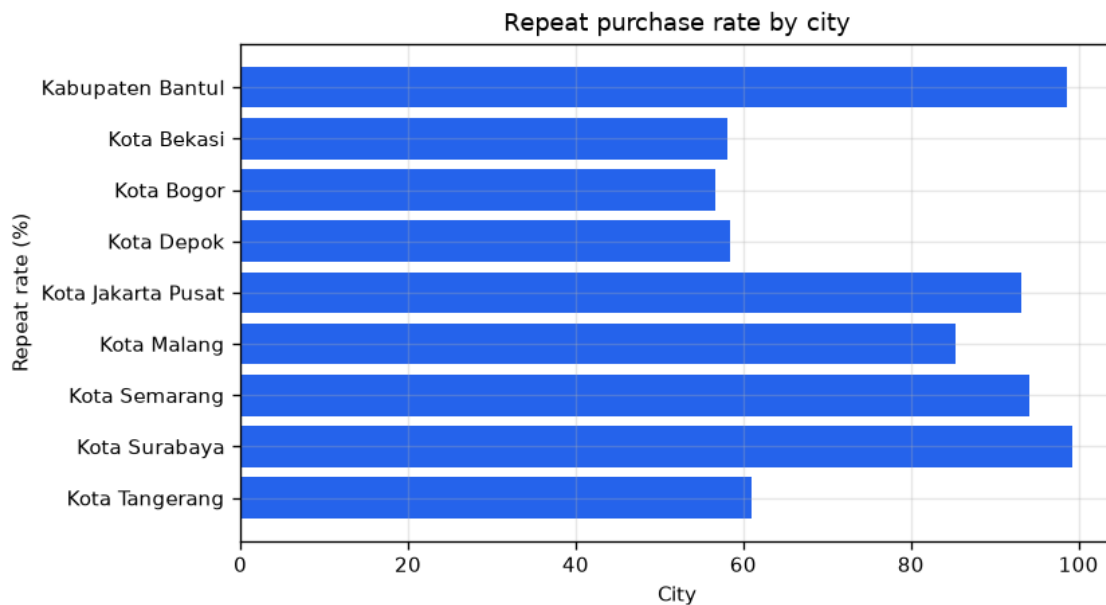
Volume rises markedly during the Eid al-Fitr period (7-13 April 2024), consistent with the initial data exploration.

6.3 Eight Business Questions

Q1. Repeat purchase & retention by region

Chi-square test of repeat purchase vs. city: statistically significant ($p = 0.0000 < \alpha = 0.05$) (Cramer's $V = 0.390$).

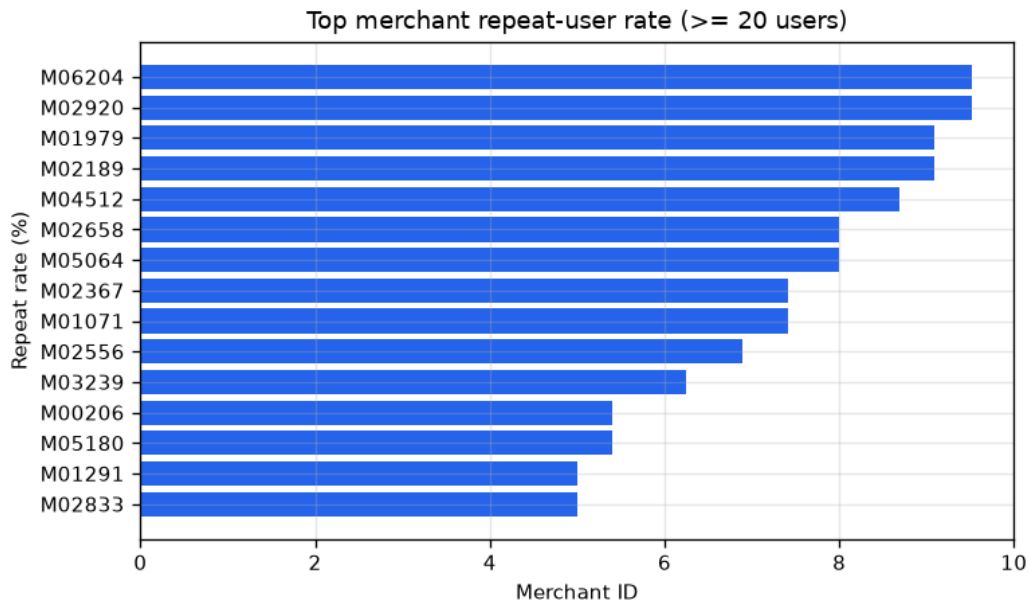
By segment: At-Risk / Dormant 60.9% ($n=3,227$), Bargain Hunters 73.0% ($n=37,961$), Champions (High-Value Loyal) 83.0% ($n=1,032$), New / Low Engagement 100.0% ($n=1$), Regular / Steady Spenders 72.7% ($n=12,197$).



Q2. Repeat orders per merchant + discount impact

Of 5,801 merchants with enough users to test, the platform-wide discount rate is 17.1%. Promo-source heuristic split: platform_baseline: 5,672, merchant_funded: 129. No promo-source column exists; 'merchant_funded' vs 'platform_baseline' is a heuristic flag from a one-sided binomial test of each merchant's discount rate against the platform-wide rate, not a ground-truth label.

Discount usage by segment: At-Risk / Dormant 16.9% ($n=7,612$), Bargain Hunters 17.1% ($n=117,415$), Champions (High-Value Loyal) 17.6% ($n=4,583$), New / Low Engagement 0.0% ($n=4$), Regular / Steady Spenders 16.8% ($n=37,383$).



Q3. Best-selling vs. rarely ordered categories

Best-selling: Food & Dining. Rarely ordered: Entertainment.

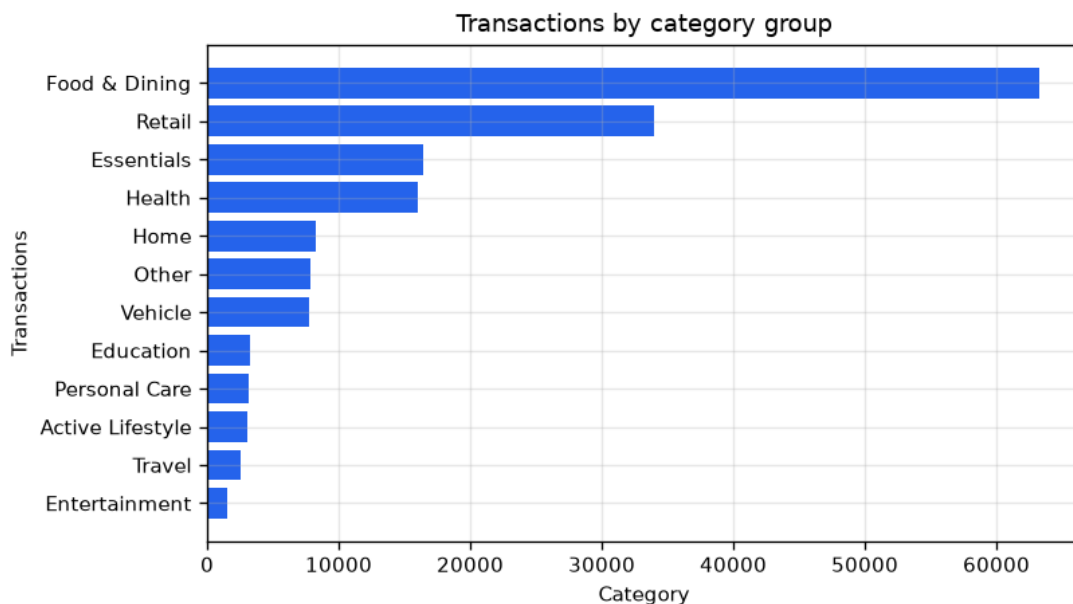
At-Risk / Dormant top categories: Food & Dining 37.4%, Retail 20.3%.

Bargain Hunters top categories: Food & Dining 38.0%, Retail 20.4%.

Champions (High-Value Loyal) top categories: Food & Dining 38.8%, Retail 19.1%.

New / Low Engagement top categories: Food & Dining 50.0%, Retail 25.0%.

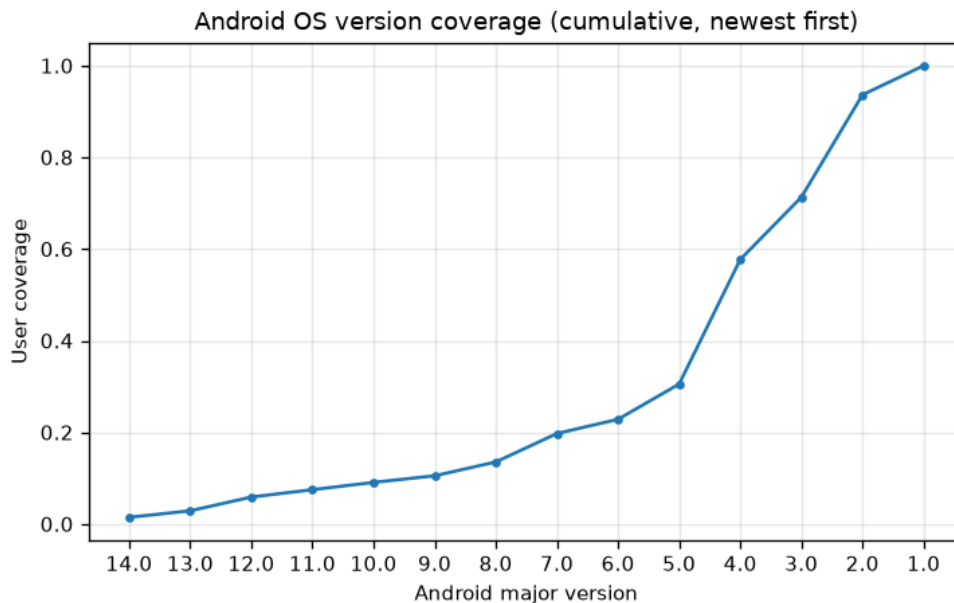
Regular / Steady Spenders top categories: Food & Dining 37.6%, Retail 20.3%.



Q4. Devices & minimum OS

Recommended minimum Android version for 90% user coverage: 1.0.

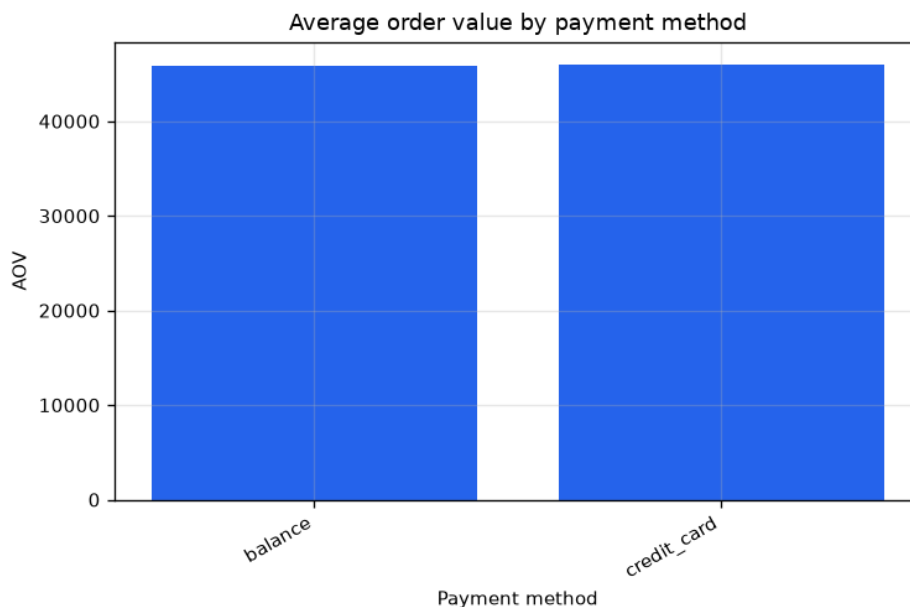
Legacy-OS share by segment: At-Risk / Dormant 68.4% (n=1,369), Bargain Hunters 86.5% (n=11,134), Champions (High-Value Loyal) 83.4% (n=373), New / Low Engagement 0.0% (n=7,871), Regular / Steady Spenders 86.4% (n=3,574).



Q5. Balance vs. credit card

Net amount distributions differ between payment methods: not statistically significant ($p = 0.8788 \geq \alpha = 0.05$) (Mann-Whitney U test). Estimated total MDR cost: 76,863,359.1.

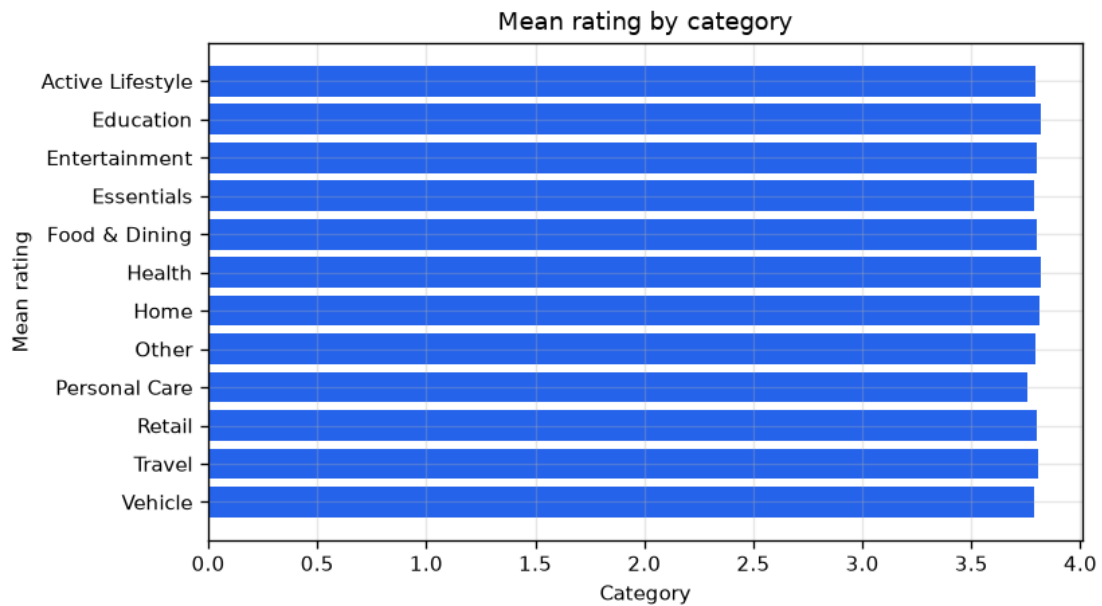
Credit-card share by segment: At-Risk / Dormant 20.2% (n=7,612), Bargain Hunters 20.1% (n=117,415), Champions (High-Value Loyal) 20.2% (n=4,583), New / Low Engagement 0.0% (n=4), Regular / Steady Spenders 20.3% (n=37,383).



Q6. Rating drivers

Refund x notes association: not statistically significant ($p = 0.9731 \geq \alpha = 0.05$) (Cramer's V = 0.000).

Mean rating by segment: At-Risk / Dormant 3.80 (n=8,019), Bargain Hunters 3.80 (n=123,698), Champions (High-Value Loyal) 3.82 (n=4,773), New / Low Engagement 3.50 (n=4), Regular / Steady Spenders 3.79 (n=39,461).

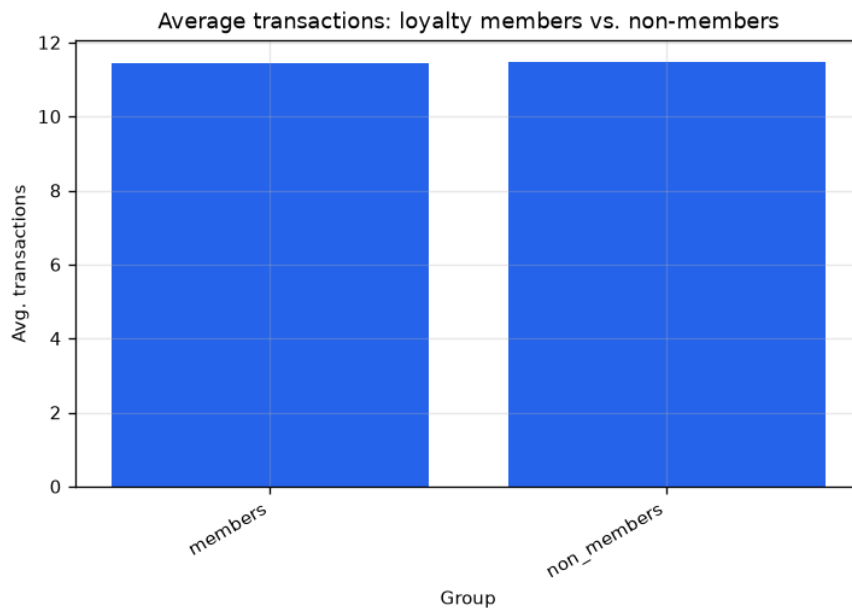


Q7. Loyalty engagement + devices

Loyalty members vs. non-members transaction frequency: not statistically significant ($p = 0.4617 \geq \alpha = 0.05$).

Segment mix among members: Regular / Steady Spenders 92.4%, At-Risk / Dormant 5.4%, Champions (High-Value Loyal) 2.2%.

Segment mix among non-members: Bargain Hunters 54.4%, New / Low Engagement 38.5%, At-Risk / Dormant 5.7%, Champions (High-Value Loyal) 1.4%.



Q8. Regions without promo usage

Never-promo share varies by city: statistically significant ($p = 0.0367 < \alpha = 0.05$) (Cramer's $V = 0.032$). A higher never-promo share can reflect promo unavailability, lower price sensitivity, or lower promo awareness in that city; this dataset cannot distinguish between the three without promo-eligibility or exposure data.

Never-promo share by segment: At-Risk / Dormant 30.3% ($n=1,078$), Bargain Hunters 17.1% ($n=11,134$), Champions (High-Value Loyal) 10.7% ($n=373$), New / Low Engagement 100.0% ($n=1$), Regular / Steady Spenders 16.6% ($n=3,574$).

